graphics cards have evolved from being used for just gaming to being used for gaming and software development and now for even AI. And it seems like with every new GPU microarchitecture, there's a lot more AI use cases that are being streamlined and incorporated into video game engines. At the core of its AI capabilities are the Tensor Cores, which are pivotal for AI processing. These Tensor Cores are specifically designed to accelerate deep learning performance, which can be seen in features like DLSS (Deep Learning Super Sampling). DLSS leverages AI to upscale lower-resolution images in real-time, enabling higher frame rates without compromising image quality.

NVIDIA's AI suite also extends to voice and image processing. NVIDIA Broadcast is a set of AI-powered tools that enhance live streams and video calls with virtual backgrounds, auto framing, and noise removal. Of late, even Windows has incorporated similar features under the collection titled Windows Studio Effects. However, these leverage any NPU that's present and are not vendor-locked like how NVIDIA's AI features tend to be.

For creators and professionals, the card's AI features translate to faster content creation and simulation. AI-assisted editing software can utilise the GPU to expedite workflows, such as automatic image tagging, object recognition, and even advanced video effects that traditionally would take hours to render. Some of the plugins that are being developed for popular software such as Adobe Premiere Pro can save a lot of time that would have otherwise been spent on tedious tasks.

Having a lot of memory on the GPU allows for local AI models to be stored and that's going to open up a lot more use cases in the near future. Aside from using AI to improve visual fidelity for upscaled or generated frames, we're now seeing how AI is making its way into interactive elements within video games. AI-driven game development tools can use the cards like the RTX 4070 SUPER for tasks such as procedural generation, NPC behaviour, and realistic physics simulations, thus enhancing the gaming experience. At CES 2024, NVIDIA showcased how far NVIDIA ACE had come along. NVIDIA ACE is a stack of tools that allows NPCs (Non Player Characters) within video games to come up with dialogues on the fly while taking into consideration real-time changes within the game. And the faces of the NPCs along with the bodies will move and emote according to the dialogue being spoken. Exciting times are ahead!

PERFORMANCE

The card is being compared against only the recently re-tested graphics cards and not any of the older graphics cards that are no longer available with us. Here's the rig it was tested on.

TEST RIG

- **Processor** – AMD Ryzen 9 7950X
- **CPU Cooler** – Corsair H115i RGB PLATINUM
- **Motherboard** – ASUS ROG CROSSHAIR X670E HERO
- **RAM** – 2x 16 GB Kingston Renegade FURY 6000 MT/s (Set to 5200 MT/s)
- **SSD** – WD Black SN850X 2 TB NVMe
- **PSU** – Cooler Master MWE 850 V2 Gold

Raster Gaming Performance

We're starting off with gaming performance in video games without RTX enabled. All of these FPS numbers are at 1440p resolution. The RTX 4070 Super scores about 111 FPS in Assassin's Creed Valhalla, 213 FPS in Battlefield 2042, 170 FPS in Metro Exodus, 142 FPS in Far Cry 6, 212 FPS in The Witcher 3 and 229 FPS in F1 2022.

Switching on to RTX performance in synthetic benchmarks as well as video games we see a close similarity to the RTX 4070 Ti across most benchmarks with barely any difference in performance. For example, in the RTX titles that we tested, the NVIDIA GeForce RTX scores about 46 FPS in CyberPunk 2077, 89 FPS in F1 2022 and 166 FPS in HITMAN 3.

Peak temperatures of the RTX 4070 Super were mostly hovering around 63-64 degrees Celsius during gaming. Only the intensive synthetic benchmarks were able to take it up any further. Every card is hovering around 70 degrees Celsius under gaming load so there's nothing of note to talk about here.

PRICE AND AVAILABILITY

In India, the card is priced at INR 63,000 inclusive of all taxes. The official launch MSRP for the RTX 4070 Super is USD 599 which is the same as the RTX 4070 so you can see why the RTX 4070 Ti has no reason to exist considering how close the RTX 4070 Super is to the 4070 Ti.

VERDICT

Honestly, I'm quite surprised that NVIDIA decided to price the NVIDIA RTX 4070 Super at USD 599. Compared to the now EOL RTX 4070 Ti, the Super easily becomes the better buy and at the same time, makes the RTX 4080 seem a little better off. That being said, even the RTX 4080 is now EOL, making the RTX 4080 Super the only 80-class card. So for folks in India, the RTX 4070 Ti is a great buy if you're looking at 1440p gaming and even some 4K gaming if you can tweak the right settings. As always, what remains to be seen is if the retail cards in India follow a similar pricing. And we all know how that pans out.

–Mithun Mohandas

SPECIFICATIONS

Code Name: AD104-350| Graphics Processing Cluster: 5 | Texture Processing Cluster: 28 | Streaming Multiprocessors: 56 | CUDA Cores: 7168 | Shader FLOPS: 36 | Tensor Cores: 224 | AI TOPS: 568 | RT Cores: 56 | RT FLOPS: 82.1 | Texture Units: 224 | ROP Units: 80 | Base Clock: 1980 MHz | Boost Clock: 2475 MHz | Memory Clock: 1313 MHz | Memory Data Rate: 21 Gbps | L1 Data Size per SM: 128 KB | L2 Cache Size: 48 MB | Total Video Memory: 12 GB | Video Memory Type: GDDR6X | Memory Interface: 192-bit | Total Memory Bandwidth: 504 GB/s | Process Node: TSMC 4N | Total Graphics Power: 220 W

CONTACT

Phone: NA | Email: Web Form | Website: https://www.nvidia.com/en-in/